

Chapter 2: Non-Parametric Procedures

Modelling Survival Data in Medical Research

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Part 3

1. Comparing Two Groups of Survival Data
2. Comparing Three or More Groups
3. Stratified Tests and Test for Trend
4. Looking Ahead
5. Q&A

Comparing Two Groups — The Question

Example 2.11 — Breast cancer (HPA staining)

29 women, divided into:

- ▶ **13** with **negative** HPA staining (appear to live longer)
- ▶ **16** with **positive** HPA staining (appear to die sooner)

The key question

Is the difference in survival between the two groups **real** (a true biological difference), or just due to **random chance**?

With only 29 patients, could we have gotten this result simply by luck, even if there was no real difference?

This is exactly what a **hypothesis test** answers.

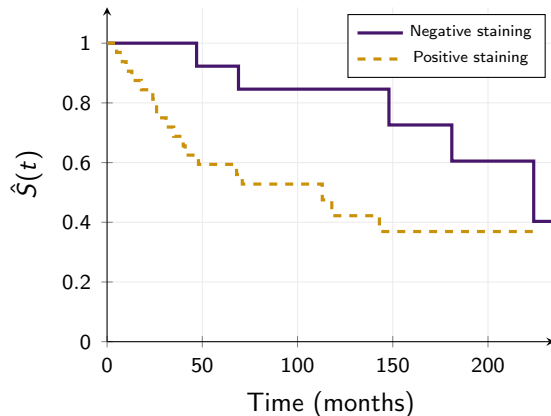
Overlaying KM Curves for Two Groups

What to look for:

- ▶ Do the curves **separate** clearly?
- ▶ Which group has higher $\hat{S}(t)$ at every time?
- ▶ Do the curves **cross**? (Crossing is important — it suggests the hazards are not proportional)
- ▶ How wide are the confidence bands?

Breast cancer

Negative $\hat{S}(t)$ stays above positive throughout follow-up. Curves do not meaningfully cross $\Rightarrow$ proportional hazards is plausible.



Hypothesis Testing — The Concept

Step 1: State the null hypothesis H_0

H_0 : There is **no difference** in survival between the two staining groups.
Any observed difference is just due to random chance.

Step 2: Compute a test statistic

A number that measures how far the data depart from H_0 .
Larger value = more evidence against H_0 .

Step 3: Compute the P -value

P -value = probability of getting a test statistic at least this large, *if H_0 were true*.

Small P -value $\Rightarrow$ the data are unlikely under $H_0 \Rightarrow$ evidence against H_0 .

Large P -value $\Rightarrow$ the data are consistent with $H_0 \Rightarrow$ no evidence against H_0 .

Interpreting a P -Value

Evidence against H_0 increases as the P -value decreases



Common misconception

A P -value is **not** the probability that H_0 is true.

It is the probability of observing results at least as extreme as those obtained, *assuming that H_0 is true*.

Good practice

Report the actual P -value and the effect size whenever possible.

A result with $P = 0.049$ is not meaningfully different from one with $P = 0.051$.

The Log-Rank Test — Building Intuition

Core idea

At each death time $t_{(j)}$, ask: given the total number of deaths d_j across both groups, how many deaths *would we expect* in Group I if survival were the same in both groups?

Expected deaths in Group I at $t_{(j)}$

$$e_{1j} = \frac{n_{1j}}{n_j} \times d_j$$

Group I's *share* of the total deaths, proportional to their share of the risk set.

Example at $t_{(j)} = 5$ months

$n_{1j} = 13$ (negative), $n_{2j} = 32$ (positive),
 $d_j = 1$ total death.

$$e_{1j} = (13/45) \times 1 = 0.289$$

We expected 0.289 deaths in the negative group but observed 0.

Consistently fewer deaths than expected $\Rightarrow$ surviving longer. Consistently more $\Rightarrow$ dying sooner.

The Log-Rank Test — The Statistic

Log-rank statistic: sum of (observed – expected)

$$U_L = \sum_{j=1}^r (d_{1j} - e_{1j}), \quad V_L = \sum_{j=1}^r v_{1j}$$

where $v_{1j} = \frac{n_{1j} n_{2j} d_j (n_j - d_j)}{n_j^2 (n_j - 1)}$ is the variance at time $t_{(j)}$.

Test statistic

$$W_L = \frac{U_L^2}{V_L} \approx \chi_1^2 \quad \text{under } H_0$$

Under H_0 : $U_L \approx 0$. Large $|U_L| \Rightarrow$ large $W_L \Rightarrow$ evidence against H_0 .

Also known as: Mantel-Cox, Peto-Mantel-Haenszel. Same test — different name by field.

Example 2.12 — Breast Cancer: Log-Rank Result

After computing 25 death-time tables (Table 2.8)

Group I (Negative staining)	
Total observed deaths:	$\sum d_{1j} = 5$
Total expected deaths:	$\sum e_{1j} = 9.565$
$U_L = 5 - 9.565$	$= -4.565$
$V_L = \sum v_{1j}$	$= 5.929$
$W_L = (-4.565)^2 / 5.929$	$= 3.515$
P -value (from χ_1^2 table)	$= 0.061$

$P = 0.061$: slight evidence against H_0 . Negative group: 5 deaths observed vs. 9.6 expected — they survived longer. $P > 0.05$: not significant at the 5% level.

The Wilcoxon Test

Same idea, different weights

$$U_W = \sum_{j=1}^r n_j \cdot (d_{1j} - e_{1j}), \quad V_W = \sum_{j=1}^r n_j^2 v_{1j}$$

Each difference $(d_{1j} - e_{1j})$ is **weighted by** n_j — the number at risk.

Effect of the weights

Early times: n_j is large $\Rightarrow$ large weight

Late times: n_j is small $\Rightarrow$ small weight

$\Rightarrow$ **Wilcoxon is more sensitive to early differences** between groups

Limitation

Performs poorly when censoring **differs between groups**.

Use **Peto-Peto** instead.

The Peto-Peto Test

Better weights: use the overall KM estimate

$$U_P = \sum_{j=1}^r \hat{S}_{KM}(t_{(j)}) \cdot (d_{1j} - e_{1j})$$

Weight each difference by the **combined** KM estimate at that time (using data from both groups).

Why is this better?

$\hat{S}_{KM}(t)$ decreases smoothly over time and is less affected by differential censoring than n_j .
In addition, it gives **less weight** to late times where there is more uncertainty.

Example 2.13 — Breast cancer

Log-rank: $P = 0.061$. Wilcoxon: $P = 0.041$. Peto-Peto: $P = 0.043$.

All three agree on direction; Wilcoxon and Peto-Peto are significant at 5%.

All Three Tests Side by Side

Test	Weights	Best when	Breast cancer P
Log-rank	None (equal)	Proportional hazards	0.061
Wilcoxon	n_j	Early differences matter	0.041
Peto-Peto	$\hat{S}_{KM}(t_{(j)})$	Unequal censoring	0.043

Decision rule: which test to use?

- 1 Plot the KM curves. Do they cross? $\Rightarrow$ use Peto-Peto or Wilcoxon
- 2 Curves do not cross (hazards likely proportional)? $\Rightarrow$ use log-rank (default)
- 3 Censoring very different between groups? $\Rightarrow$ use Peto-Peto over Wilcoxon

Default choice: **log-rank test**, unless proportional hazards is in doubt.

What Is the Proportional Hazards Assumption?

Definition

$h_1(t) = \psi \cdot h_2(t)$ for all $t \geq 0$: the hazard *ratio* is constant over time.

Consequence for $S(t)$

$$S_1(t) = \{S_2(t)\}^\psi$$

The survivor functions **never cross**.

$\psi < 1$: Group I always better.

$\psi > 1$: Group I always worse.

How to check informally

- 1 Plot both KM curves
- 2 **No crossing**: PH plausible
- 3 **Crossing**: PH likely violated

Non-crossing is necessary but not sufficient.

If KM curves do not cross, use the **log-rank test**.

Comparing More Than Two Groups

Suppose we have $g \geq 3$ groups (e.g. three age groups: 21–40, 41–60, 61+). We want to test H_0 : all groups have the same survival distribution.

Extension of the log-rank test

For each group $k = 1, \dots, g - 1$, compute:

$$U_{Lk} = \sum_{j=1}^r \left(d_{kj} - \frac{n_{kj} d_j}{n_j} \right)$$

Collect these into a vector $\mathbf{U}_L$ and a variance-covariance matrix $\mathbf{V}_L$.

Test statistic: $W = \mathbf{U}_L^\top \mathbf{V}_L^{-1} \mathbf{U}_L \sim \chi_{g-1}^2$ under H_0 .

Degrees of freedom

For 3 groups: χ_2^2 , 4 groups: χ_3^2 . Statistical software computes this automatically.

Stratified Log-Rank Test — Why?

The problem: confounding

Sometimes we want to compare two treatments, but the patients differ in another important variable (e.g. age group, hospital centre, disease stage).

If we ignore this **stratifying variable**, the comparison may be distorted.

Example 2.15 — Melanoma trial

Two immunotherapy treatments (BCG vs. *C. parvum*) across three age groups (21–40, 41–60, 61+).

We want to test for a treatment difference *after accounting for age group*.

Stratified Log-Rank — Solution

Strategy

Run the log-rank test **separately within each stratum** (age group), then **combine** the evidence across strata.

Why not just run one big log-rank test ignoring age?

Older patients may have worse survival overall, regardless of treatment.

Comparing treatments without adjusting for age could falsely attribute age effects to the treatment.

Stratification **removes** the age effect before comparing treatments.

Stratification is one of the simplest and most important tools for controlling confounding in survival analysis.

Stratified Log-Rank — Formula

Within each stratum k

Compute U_{Lk} and V_{Lk} using the standard log-rank formulas, but only using data from stratum k .

Combine across all s strata

$$W_S = \frac{\left(\sum_{k=1}^s U_{Lk} \right)^2}{\sum_{k=1}^s V_{Lk}} \sim \chi_1^2 \quad \text{under } H_0$$

Example 2.15 result

$W_S = 1.237^2 / 2.225 = 0.688$, $P > 0.1$ — no significant treatment difference.
Unstratified: $W_L = 0.756$ — almost identical, age adjustment not critical here.

Log-Rank Test for Trend

When groups are *ordered*

Sometimes the g groups have a natural order: increasing dose, disease stage, age group.

A general χ^2_{g-1} test might miss a clear trend.

The **trend test** uses the ordering and has only χ^2_1 — more power.

Trend statistic

Assign codes w_k (e.g. $-1, 0, 1$ for three ordered groups). Then:

$$U_T = \sum_{k=1}^g w_k (d_{k\cdot} - e_{k\cdot}), \quad V_T = \sum_{k=1}^g (w_k - \bar{w})^2 e_{k\cdot}.$$

$W_T = U_T^2 / V_T \sim \chi^2_1$ — one degree of freedom regardless of g .

Trend Test vs. General Test — When to Use Which

General χ^2_{g-1} test

- ▶ Groups have **no natural ordering**
- ▶ Or the relationship is not expected to be monotone
- ▶ Tests any pattern of difference
- ▶ Uses $g - 1$ degrees of freedom

Trend test (χ^2_1)

- ▶ Groups are **ordered** (dose, stage, age)
- ▶ A monotone trend is expected
- ▶ More powerful for detecting a gradient
- ▶ Uses only 1 degree of freedom

A trend test has more power to detect a *monotone* difference because it uses the information about *ordering* of the groups.

Estimating $S(t)$

- ▶ **Empirical**: simple fraction — no censoring only
- ▶ **Life-table**: grouped intervals — actuarial assumption
- ▶ **Kaplan-Meier** *: one interval per death — standard method
- ▶ **Nelson-Aalen**: via $\tilde{H}(t) = \sum d_j/n_j$ — better for small n

Precision and summaries

- ▶ Greenwood's formula: $\widehat{\text{se}}\{\hat{S}(t)\}$
- ▶ Pointwise 95% CI for $S(t)$
- ▶ Median: smallest t with $\hat{S}(t) < 0.5$
- ▶ SIQR: half the interquartile range

Two-group tests

- ▶ **Log-rank:** best under proportional hazards; the default
- ▶ **Wilcoxon:** more sensitive to early differences
- ▶ **Peto-Peto:** robust to unequal censoring between groups

Extensions

- ▶ **g groups:** χ^2_{g-1} test statistic
- ▶ **Stratified log-rank:** adjusts for a confounding variable
- ▶ **Trend test:** ordered groups, uses only χ^2_1

The **Kaplan-Meier estimate** and the **log-rank test** are the two most important tools in Chapter 2.

What we can do now:

- ✓ Estimate $S(t)$ with KM
- ✓ Find the median survival time
- ✓ Draw KM curves with CI bands
- ✓ Test if two groups differ (log-rank)
- ✓ Handle censoring correctly

What we cannot do yet:

- ✗ Adjust for multiple covariates at once (e.g. age AND sex AND tumour size)
- ✗ Estimate how much a risk factor changes survival
- ✗ Predict survival for a new patient

Chapter 3: Cox Proportional Hazards Model

The **Cox model** is the backbone of modern survival analysis. It lets us compare individuals moving through time, while accounting for many risk factors at once. At its core, it translates complex covariate effects into **hazard ratios**, telling us who is at higher or lower risk, and by how much.

Questions?

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